ATHUL KRISHNA R

AI/ML Developer | Python | Full Stack Developer

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PROFESSIONAL SUMMARY

MCA student with hands-on experience building full-stack applications and AI-powered solutions.
Skilled in Python, FastAPI, PyTorch, OpenCV, AI/ML, RAG, agentic AI, REST APIs, React, Node.js, and MongoDB.
Built AgriPulse (AgriMitra 360), a multimodal smart agriculture platform combining computer vision, RAG,
agentic AI, API integrations, and Malayalam voice interaction. Passionate about solving real-world problems.

TECHNICAL SKILLS

Programming & AI: Python, JavaScript, AI/ML, PyTorch, Torchvision, OpenCV, MobileNet
AI & Automation: RAG, LLM concepts, Agentic AI, ReAct, Tool Calling, AI-assisted Development
Backend & APIs: FastAPI, Node.js, Express.js, REST APIs, API Integration, CRUD, Socket.IO
Frontend: React.js, Vite, Tailwind CSS, HTML5, CSS3, Bootstrap, Responsive Web Design
Databases & Tools: MongoDB Atlas, Mongoose, SQLite | Git, GitHub, Postman, VS Code

FEATURED PROJECTS

AgriPulse (AgriMitra 360) - Multimodal Smart Agriculture Platform | AI / Agentic AI

Tech: Python, FastAPI, PyTorch, Torchvision, MobileNet, OpenCV, RAG, ReAct Agent, Tool Calling, React, Node
- Built AI agriculture platform combining PyTorch/MobileNet leaf scanner and OpenCV HSV lesion analysis.
- Developed ReAct agent with multi-tool calling for mandi prices, weather spray safety, and KAU/ICAR advisory.
- Integrated bilingual Malayalam/English voice interaction via Web Speech API and FastAPI AI microservice.

Ey Auto - Auto Rickshaw Booking Platform | B.Sc Final Year Project

Tech: React.js, Node.js, Express.js, MongoDB, Socket.IO
- Engineered ride-booking platform with Passenger, Driver, and Admin modules, RBAC auth, and live Socket.IO.

POS Management System - Desktop Application

Tech: Electron.js, React.js, Node.js, SQLite
- Standalone desktop retail POS solution with offline-first SQLite database, billing, and inventory tracking.

EcoClean - AI-Powered Smart Waste Management Portal | AI / Full Stack

Tech: MERN Stack, Google Gemini Vision API, React.js, Node.js, Express, MongoDB, Leaflet.js
- Waste reporting platform classifying images via Gemini Vision with Leaflet mapping and Eco-Points gamification.

EDUCATION & CERTIFICATIONS

Master of Computer Applications (MCA) | MEA Engineering College, Perinthalmanna (2025 - 2027)
Bachelor of Science in Computer Science | KRS Sree Narayana College, Valanchery (2022 - 2025)
Full Stack Development (MERN Stack) - Certified User by Illinois Institute of Technology & Entri Elevate
Full Stack Web Development (MERN) with Capstone Project - Certified by Entri Elevate
8-Month Intensive Full Stack Development Program focusing on modern web standards & desktop environments